Topics 7.6 – 7.10: MCQ Practice

	Chemical Equation	K_p
#1	$2\ XY(g) \rightleftarrows X_2(g) + Y_2(g)$	0.16
#2	$\frac{1}{2}\ X_2(g) + \frac{1}{2}\ Y_2(g) \rightleftarrows XY(g)$	?

1. Based on the information above, the value of K_p for Equation #2 is closest to

   (A) 0.40 (B) 2.5 (C) 3.1 (D) 6.3

	Chemical Equation	K_c
#1	$PbF_2(s) \rightleftarrows Pb^{2+}(aq) + 2\ F^-(aq)$	3.6×10^{-8}
#2	$HF(aq) \rightleftarrows H^+(aq) + F^-(aq)$	6.8×10^{-4}
#3	$PbF_2(s) + 2\ H^+(aq) \rightleftarrows Pb^{2+}(aq) + 2\ HF(aq)$	?

2. Based on the information above, the value of K_c for Equation #3 is closest to

   (A) 2.4×10^{-11}

   (B) 5.3×10^{-5}

   (C) 1.1×10^{-4}

   (D) 7.8×10^{-2}

$$2\ H_2S(g) + CH_4(g) \rightleftarrows CS_2(g) + 4\ H_2(g) \qquad K_c = 3.4 \times 10^{-4}$$

3. A 0.10 mol sample of each of the four species in the reaction represented above is injected into a rigid, previously evacuated 1.0 L container. Which of the following species will have the highest concentration when the system reaches equilibrium?

   (A) $H_2S(g)$

   (B) $CH_4(g)$

   (C) $CS_2(g)$

   (D) $H_2(g)$

$$H_2(g) + I_2(g) \rightleftarrows 2\ HI(g) \qquad K_c = 51 \text{ at } 450°C$$

4. At 450°C, 2.0 moles each of $H_2(g)$, $I_2(g)$, and $HI(g)$ are combined in a 1.0 L rigid container. Which of the following will occur as the system moves toward equilibrium as represented by the equation above?

   (A) More $H_2(g)$ and $I_2(g)$ will be produced because $Q_c > K_c$.

   (B) More $HI(g)$ will be produced because $Q_c < K_c$.

   (C) The total pressure in the container will decrease because $Q_c < K_c$.

   (D) No net reaction will occur, because the number of molecules is the same on both sides of the equation.

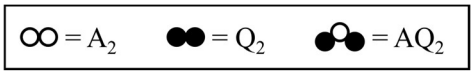
$$X_2(g) + 3\ Y_2(g) \rightleftarrows 2\ XY_3(g) \qquad K_c = 0.40$$

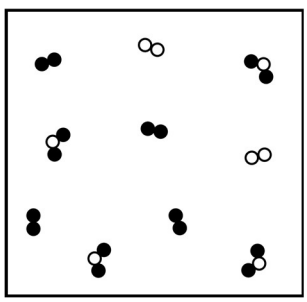
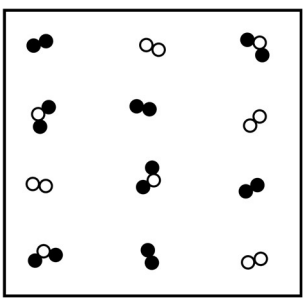
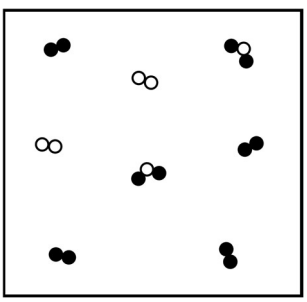
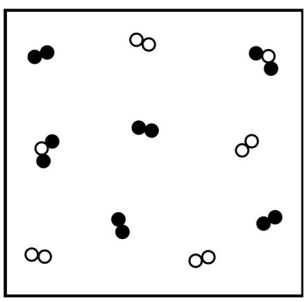
5. A mixture of $X_2(g)$, $Y_2(g)$, and $XY(g)$ is placed in a previously evacuated, rigid container and allowed to reach equilibrium at a constant temperature, as shown above. Which of the following sets of initial concentrations would lead to the formation of more $XY_3(g)$ as the system moves toward equilibrium?

	$[X_2]_{initial}$	$[Y_2]_{initial}$	$[XY_3]_{initial}$
(A)	0.50 *M*	1.5 *M*	2.0 *M*
(B)	0.50 *M*	1.5 *M*	4.0 *M*
(C)	1.0 *M*	3.0 *M*	2.0 *M*
(D)	1.0 *M*	3.0 *M*	4.0 *M*

$$A_2(g) + 2\ Q_2(g) \rightleftarrows 2\ AQ_2(g) \qquad K_p = 0.125 \text{ at } 500 \text{ K}$$

6. Substance $A_2(g)$ reacts with substance $Q_2(g)$ to produce substance $AQ_2(g)$ according to the equation above. Which of the following particulate diagrams is the best representation of the reaction mixture after equilibrium has been achieved at 500 K? Assume that each molecule in the diagram represents 1 atm of gas pressure.



(A)	(B)	(C)	(D)
			

$$CO(g) + H_2O(g) \rightleftarrows CO_2(g) + H_2(g)$$

7. Which of the following changes will increase the amount of $CO_2(g)$ in the equilibrium system represented by the equation above?

   (A) Removing $H_2(g)$ from the system

   (B) Removing $H_2O(g)$ from the system

   (C) Increasing the volume of the reaction vessel

   (D) Decreasing the volume of the reaction vessel

Reaction	Chemical Equation
A	$H_2(g) + Br_2(g) \rightleftarrows 2\ HBr(g)$
B	$N_2(g) + 3\ H_2(g) \rightleftarrows 2\ NH_3(g)$
C	$2\ H_2O_2(g) \rightleftarrows 2\ H_2O(g) + O_2(g)$
D	$N_2O_4(g) \rightleftarrows 2\ NO_2(g)$

8. The reactions represented above are carried out in sealed, rigid containers and allowed to reach equilibrium. If the volume of each container is reduced from 1.0 L to 0.5 L at constant temperature, for which of the reactions will the amount of product(s) be increased?

   (A) Reaction A (B) Reaction B (C) Reaction C (D) Reaction D

$$PCl_5(g) \rightleftarrows PCl_3(g) + Cl_2(g)$$

9. A pure sample of $PCl_5(g)$ is placed in a previously evacuated 2.0 L reaction vessel. When the system has reached equilibrium at 500 K according to the equation represented above, the total pressure of the reaction mixture is equal to 1.5 atm. Then the volume of the reaction vessel is reduced from 2.0 L to 1.0 L, and equilibrium is reestablished for the reaction system at 500 K. At this point, the total pressure in the reaction vessel should be

   (A) Less than 1.5 atm

   (B) Greater than 1.5 atm but less than 3.0 atm

   (C) Equal to 3.0 atm

   (D) Greater than 3.0 atm

$$CO(g) + 2\,H_2(g) \rightleftarrows CH_3OH(g) \qquad \Delta H < 0$$

10. The synthesis of $CH_3OH(g)$ from $CO(g)$ and $H_2(g)$ is represented by the equation above. The value of the equilibrium constant, K_c, for the reaction at 483 K is 14.5.

    When the temperature of the system is increased to 650 K, which of the following identifies what happens to the value of K_c and provides the correct justification?

	K_c	Justification
(A)	K_c will decrease	The reaction is exothermic.
(B)	K_c will decrease	The activation energy of the reverse reaction is decreased.
(C)	K_c will increase	The reaction is exothermic.
(D)	K_c will increase	The activation energy of the forward reaction is decreased.

$$N_2O_4(g) \rightleftarrows 2\,NO_2(g) \qquad \Delta H > 0$$

11. A mixture of $N_2O_4(g)$ and $NO_2(g)$ is placed in a container and allowed to reach equilibrium according to the equation shown above. The amount of $NO_2(g)$ present in the reaction mixture at equilibrium can be maximized by

    (A) increasing the temperature and decreasing the volume of the container

    (B) increasing the temperature and increasing the volume of the container

    (C) decreasing the temperature and decreasing the volume of the container

    (D) decreasing the temperature and increasing the volume of the container

$$2\,CO(g) \rightleftarrows C(s) + CO_2(g)$$

12. Which of the following changes will increase the amount of $CO_2(g)$ in the reaction system at equilibrium represented above?

    (A) Adding a suitable catalyst to the reaction mixture

    (B) Removing a small amount of $C(s)$ from the reaction vessel

    (C) Increasing the pressure by adding $Ne(g)$ to the system at constant temperature

    (D) Decreasing the volume of the reaction vessel at constant temperature